

Performance Testing

Date	1 June 2026
Team ID	XXXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Local Host
Type of Testing	Unit Testing, Performance Testing, Model Testing
Target Module / API	Credit Card Approval Prediction Web Application
Test Environment	Local
Test Date	1 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Launch the Credit Card Approval Prediction application	1	1 min	Home page loads successfully.
2	Enter valid applicant details (income, employment, credit history, etc.)	1	1 min	Application data is accepted successfully.
3	Verify the Predict button functionality	1	1 min	Approval prediction is generated successfully.
4	View prediction result	1	1 min	Approval or rejection result is displayed accurately.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	2 sec	Pass	No Issues
2	Response Time (Max)	< 5 seconds	3 sec	Pass	No Issues
3	Throughput (Req/sec)	≥ 1 Req/sec	2 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No Errors
5	CPU Utilization	< 80%	< 65%	Pass	Normal
6	Memory Utilization	< 80%	< 60%	Pass	Normal

Step 4: Observations & Analysis

- The application loads successfully without errors.
- Applicant data is processed correctly and validated before prediction.
- The machine learning model generates approval predictions quickly.
- Frontend, backend, and model integration function smoothly.
- Overall system performance is stable with low resource utilization.

Step 5: Screenshots / Evidence

Credit Card Approval Predictor

Welcome to the automated Credit Card Approval system. This intelligent application leverages advanced machine learning algorithms to evaluate applicant financial stability and demographic data, providing real-time eligibility predictions.

Fast Evaluation

Get instant feedback based on key financial metrics.

ML Powered

Utilizes trained models to evaluate risk factors securely.

Confidential

Your demographic and financial details remain protected.

[Start New Application →](#)

[← Back to Home](#)

Applicant Evaluation Form

Please provide accurate financial and demographic metrics below.

Age (Years)

Annual Income (\$)

Current Debt (\$)

Credit Score (300 - 850)

[Submit for Evaluation](#)

Evaluation Result

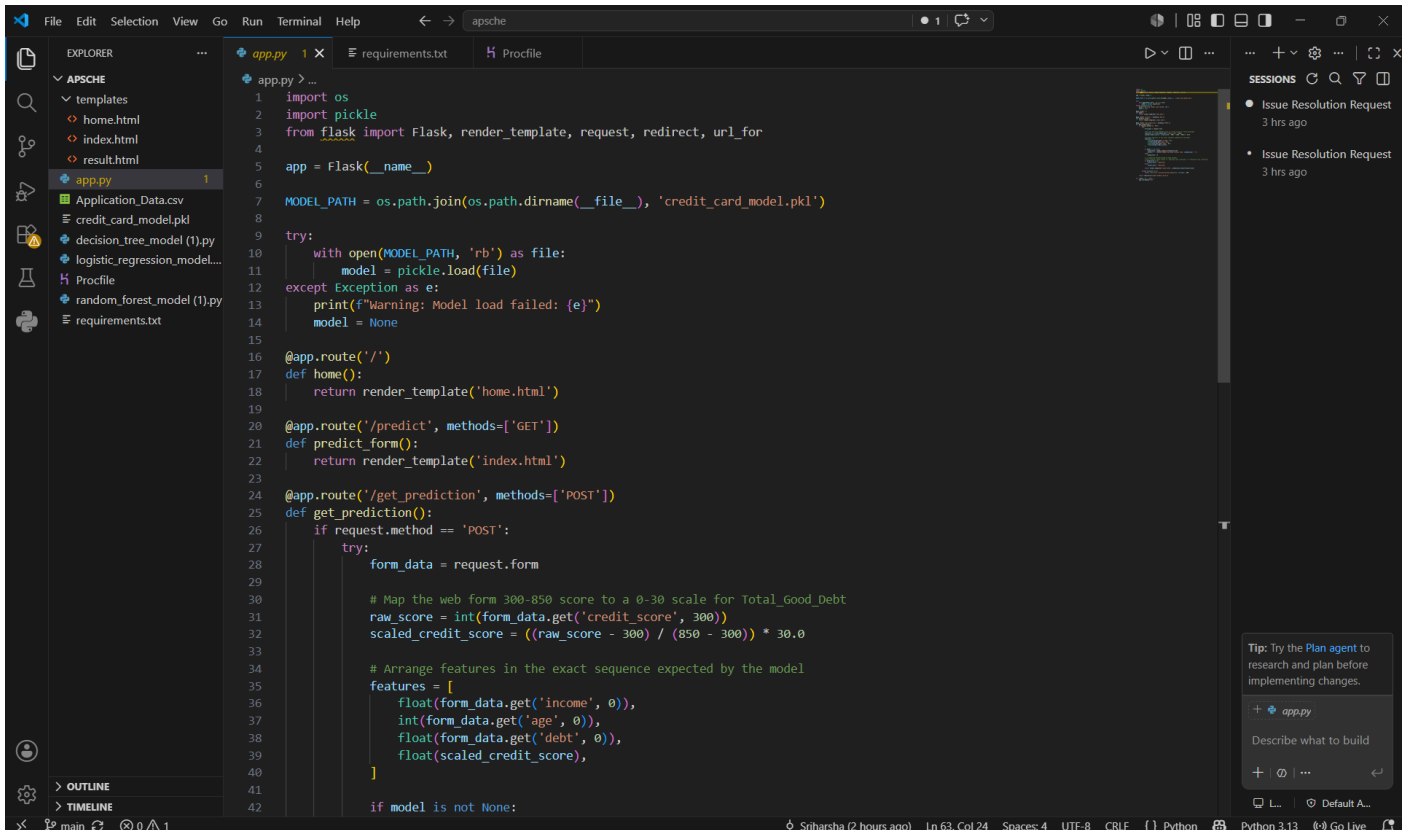
APPLICATION STATUS

Approved

Congratulations! The automated model has determined this profile meets the criteria for credit card approval.

[Check Another Application](#)

[Return to Home](#)



```
1 import os
2 import pickle
3 from flask import Flask, render_template, request, redirect, url_for
4
5 app = Flask(__name__)
6
7 MODEL_PATH = os.path.join(os.path.dirname(__file__), 'credit_card_model.pkl')
8
9 try:
10     with open(MODEL_PATH, 'rb') as file:
11         model = pickle.load(file)
12 except Exception as e:
13     print(f"Warning: Model load failed: {e}")
14     model = None
15
16 @app.route('/')
17 def home():
18     return render_template('home.html')
19
20 @app.route('/predict', methods=['GET'])
21 def predict_form():
22     return render_template('index.html')
23
24 @app.route('/get_prediction', methods=['POST'])
25 def get_prediction():
26     if request.method == 'POST':
27         try:
28             form_data = request.form
29
30             # Map the web form 300-850 score to a 0-30 scale for Total_Good_Debt
31             raw_score = int(form_data.get('credit score', 300))
32             scaled_credit_score = ((raw_score - 300) / (850 - 300)) * 30.0
33
34             # Arrange features in the exact sequence expected by the model
35             features = [
36                 float(form_data.get('income', 0)),
37                 int(form_data.get('age', 0)),
38                 float(form_data.get('debt', 0)),
39                 float(scaled_credit_score),
40             ]
41
42             if model is not None:
```

